

An Evidence Qualification Framework for Annotation-Level Formula Discrimination in Traditional Chinese Medicine

Supplementary Information

The supplement reports the unified instrument, cross-formula portability benchmark, original and sensitivity agreement analyses, and machine-readable audit outputs.

Supplementary Table S1A | Formula identity and dosing

The verified total daily crude mass was 361 g.

Formula position	Chinese name	Herb pinyin	Pharmacopoeia name	Medicinal part	Processing state	Daily dose g
1	南沙参	Nanshashen	Adenophorae Radix	root	unprocessed	30
2	白术	Baizhu	Atractylodis Macrocephalae Rhizoma	rhizome	unprocessed	30
3	茯苓	Fuling	Poria	sclerotium	unprocessed	15
4	甘草	Gancao	Glycyrrhizae Radix et Rhizoma	root and rhizome	unprocessed	6
5	陈皮	Chenpi	Citri Reticulatae Pericarpium	pericarp	aged dried pericarp	15
6	法半夏	Banxia	Pinelliae Rhizoma Praeparatum	tuber	processed Pinellia rhizome (Fa Banxia)	15
7	北柴胡	Chaihu	Bupleuri Radix	root	unprocessed	15
8	白芍	Baishao	Paeoniae Radix Alba	root	processed white paeony root	30
9	枳壳	Zhike	Aurantii Fructus	immature fruit	unprocessed	15
10	黄芪	Huangqi	Astragali Radix	root	unprocessed	60
11	北沙参	Beishashen	Glehniae Radix	root	unprocessed	30
12	豆蔻	Doukou	Amomi Fructus	fruit	unprocessed	10
13	薏苡仁	Yiyiren	Coicis Semen	seed	unprocessed	20
14	苦杏仁	Kuxingren	Armeniacae Semen Amarum	seed	unprocessed	10
15	郁李仁	Yuliren	Pruni Semen	seed	unprocessed	30
16	柏子仁	Baiziren	Platycladi Semen	seed	unprocessed	30

Supplementary Table S1B | Taxonomy and database mapping

Identity fields are separated from dosing to prevent narrow-column fragmentation.

Formula position	Latin binomial	Database query synonyms	Database species matching	Identity evidence status
1	Adenophora stricta Miq.; Adenophora tetraphylla (Thunb.) Fisch.	南沙参; Adenophorae Radix; Nanshashen	accepted pharmacopoeial source species; database record retained only when the herb synonym resolves to the pharmacopoeial drug	locked; dose confirmed
2	Atractylodes macrocephala Koidz.	白术; Atractylodis Macrocephalae Rhizoma; Baizhu	exact pharmacopoeial species or curated herb-level synonym	locked; dose confirmed
3	Wolfiporia extensa (Peck) Ginns	茯苓; Poria; Fuling	accepted pharmacopoeial fungal source name and retained database synonym	locked; dose confirmed
4	Glycyrrhiza uralensis Fisch.; Glycyrrhiza inflata Bat.; Glycyrrhiza glabra L.	甘草; Glycyrrhizae Radix et Rhizoma; Gancao	accepted pharmacopoeial source species or curated drug-name synonym	locked; dose confirmed
5	Citrus reticulata Blanco	陈皮; Citri Reticulatae Pericarpium; Chenpi	exact pharmacopoeial species or drug-name synonym	locked; dose confirmed

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Formula position	Latin binomial	Database query synonyms	Database species matching	Identity evidence status
6	Pinellia ternata (Thunb.) Breit.	法半夏; Pinelliae Rhizoma Praeparatum; Fa Banxia; Banxia	only prepared-Pinellia synonym retained for formula identity; raw Pinelliae Rhizoma excluded	locked; dose and processing form confirmed
7	Bupleurum chinense DC.	北柴胡; 柴胡; Bupleuri Radix; Beichaihu; Chaihu	formula identity restricted to Bupleurum chinense DC.; southern Bupleurum source excluded	locked; dose and northern source confirmed
8	Paeonia lactiflora Pall.	白芍; Paeoniae Radix Alba; Baishao	exact pharmacopoeial species and processed-drug synonym	locked; dose confirmed
9	Citrus aurantium L.	枳壳; Aurantii Fructus; Zhike	exact pharmacopoeial species or curated drug-name synonym	locked; dose confirmed
10	Astragalus mongholicus Bunge; Astragalus membranaceus (Fisch.) Bunge	黄芪; Astragali Radix; Huangqi	accepted pharmacopoeial source species or curated drug-name synonym	locked; dose confirmed
11	Glehnia littoralis F. Schmidt ex Miq.	北沙参; Glehniae Radix; Beishashen	exact pharmacopoeial species or curated drug-name synonym	locked; dose confirmed
12	Amomum kravanh Pierre ex Gagnep.; Amomum compactum Sol. ex Maton	豆蔻; 豆蔻仁; Amomi Fructus; Doukou	accepted pharmacopoeial source species; fruit terminology used for the pharmacopoeial drug	locked; dose confirmed
13	Coix lacryma-jobi L. var. ma-yuen (Roman.) Stapf.	薏苡仁; Coicis Semen; Yiyiren	exact pharmacopoeial taxon or curated drug-name synonym	locked; dose confirmed
14	Prunus armeniaca L. var. ansu Maxim.; Prunus mandshurica (Maxim.) Koehne; Prunus sibirica L.	苦杏仁; Armeniacae Semen Amarum; Kuxingren	accepted pharmacopoeial source species or curated drug-name synonym	locked; dose confirmed
15	Prunus japonica Thunb.; Prunus humilis Bunge	郁李仁; Pruni Semen; Yuliren	accepted pharmacopoeial source species or curated drug-name synonym	locked; dose confirmed
16	Platycladus orientalis (L.) Franco	柏子仁; Platycladi Semen; Baiziren	exact pharmacopoeial species or curated drug-name synonym	locked; dose confirmed

Supplementary Table S2 | Unified SQRCD EQF gate profile

The categorical profile is primary; no additive score is used.

Domain	Gate role	State
Formula identity	mandatory	qualified
Preparation chemistry	mandatory	partially qualified
Chemical reconstruction	mandatory	partially qualified
Target provenance	mandatory	partially qualified
Human disease context	supportive	partially qualified
Human genetics	supportive	not estimable
Formula perturbation and task identifiability	supportive	not estimable
Structural eligibility	dependent	stopped

Supplementary Table S3 | Target-evidence tier sensitivity

Complete machine-readable values are retained in the repository.

Evidence tier	Relation rows	Unique herb gene edges	Unique compounds	Unique genes	Target FDI	Multiherb target fraction
all disease supported	458	368	80	139	0.352	0.504
ChEMBL direct quantitative le 10uM	152	138	14	29	0.215	0.793
TCMSP labelled validated	306	245	74	121	0.390	0.430
human genetic disease support	2	2	1	1	0.323	1
curated or clinical disease support	69	56	26	17	0.302	0.647
GeneCards score ge 3 support	452	362	78	136	0.352	0.500

Supplementary Table S4 | Disease-evidence source composition

Complete machine-readable values are retained in the repository.

Disease evidence source rule	Unique genes	Interpretation
Open Targets primary phenotype score >0	23	supportive annotation; not automatically causal
Open Targets genetic evidence score >0	0	supportive annotation; not automatically causal
Open Targets clinical evidence score >0	2	supportive annotation; not automatically causal
GWAS Catalog association present	1	supportive annotation; not automatically causal
GeneCards relevance score >=3	136	supportive annotation; not automatically causal
OMIM gene-map row present	15	supportive annotation; not automatically causal

Supplementary Table S5 | Complete subformula retention summary

Complete machine-readable values are retained in the repository.

N retained	N removed	N combinations	Target median	Target 5th percentile	Target 95th percentile	Compound median	Compound 5th percentile	Compound 95th percentile	Fraction target retention ≥ 0.80
1	15	16	0.122	0.005	0.390	0.078	0.033	0.194	0
2	14	120	0.302	0.028	0.526	0.161	0.079	0.292	0
3	13	560	0.410	0.122	0.605	0.236	0.136	0.368	0
4	12	1820	0.482	0.209	0.676	0.312	0.196	0.452	0.001
5	11	4368	0.547	0.302	0.734	0.382	0.256	0.526	0.007
6	10	8008	0.612	0.389	0.777	0.449	0.314	0.592	0.027
7	9	11440	0.669	0.453	0.820	0.511	0.373	0.651	0.076
8	8	12870	0.712	0.525	0.856	0.572	0.434	0.706	0.168
9	7	11440	0.755	0.583	0.892	0.630	0.492	0.756	0.306
10	6	8008	0.799	0.633	0.921	0.686	0.552	0.805	0.479
11	5	4368	0.835	0.691	0.950	0.743	0.615	0.848	0.662
12	4	1820	0.871	0.748	0.971	0.802	0.680	0.889	0.820
13	3	560	0.914	0.799	0.993	0.859	0.747	0.927	0.929
14	2	120	0.950	0.855	1	0.912	0.809	0.961	0.983
15	1	16	0.978	0.906	1	0.960	0.875	0.986	1
16	0	1	1	1	1	1	1	1	1

Supplementary Table S6 | Exact Shapley annotation contributions

Complete machine-readable values are retained in the repository.

Herb	Crude mass g	Target shapley	Target unique component	Target redundant component	Compound shapley
Nanshashen	30	0.001	0	0.001	0.042
Baizhu	30	0.003	0	0.003	0.055
Fuling	15	0	0	0	0.070
Gancao	6	0.097	0.043	0.053	0.157
Chenpi	15	0.035	0.007	0.028	0.051
Banxia	15	0.059	0.043	0.016	0.068
Chaihu	15	0.153	0.079	0.074	0.150
Baishao	30	0.042	0.022	0.020	0.089
Zhike	15	0.005	0	0.005	0.025
Huangqi	60	0.169	0.086	0.082	0.089
Beishashen	30	0.133	0.043	0.090	0.051
Doukou	10	0.071	0.022	0.050	0.046
Yiyiren	20	0.011	0.007	0.004	0.025
Kuxingren	10	0.163	0.115	0.048	0.049
Yuliren	30	0.021	0	0.021	0.005
Baiziren	30	0.038	0.029	0.009	0.027

Supplementary Table S7 | Redundancy and concentration metrics

Complete machine-readable values are retained in the repository.

Unit	Nodes	Normalized entropy	Entropy equivalent equally supported nodes	Support HHI	Support Gini	Multiherb support fraction	Mean herb support	Shapley HHI	Shapley Gini	Maximum shapley share
target	139	0.936	101.409	0.013	0.427	0.504	2.647	0.119	0.524	0.169
compound	2657	0.975	2178.164	0.001	0.260	0.203	1.425	0.089	0.343	0.157

Supplementary Table S8 | Annotation edge-thinning summary

Complete machine-readable values are retained in the repository.

Edge retention rate	Replicates	Top1 rank reversal probability	Target coverage median	Target coverage 2.5th percentile	Target coverage 97.5th percentile	Target FDI median	Target FDI 2.5th percentile	Target FDI 97.5th percentile	Herb loss rank tau median	Herb loss rank tau 2.5th percentile	Herb loss rank tau 97.5th percentile	Top3 jaccard median	Top3 jaccard 2.5th percentile	Top3 jaccard 97.5th percentile
0.600	1000	0.480	107	98	115	0.380	0.364	0.395	0.881	0.770	0.961	1	0.500	1
0.700	1000	0.392	116	108	124	0.371	0.358	0.386	0.913	0.822	0.972	1	0.500	1
0.800	1000	0.241	124	118	131	0.364	0.352	0.375	0.942	0.878	0.986	1	0.500	1
0.900	1000	0.077	132	127	136	0.357	0.349	0.366	0.967	0.919	0.995	1	1	1

Supplementary Table S9 | Degree-preserving null

Complete machine-readable values are retained in the repository.

Metric	Observed	Null mean	Null sd	Empirical upper tail p
shapley HHI	0.119	0.126	0.004	0.977
maximum shapley share	0.169	0.187	0.011	0.973
LOHO loss HHI	0.140	0.148	0.013	0.735

Supplementary Table S10 | GSE245885 quality-control summary

Complete machine-readable values are retained in the repository.

Dataset	N	N STC	N NP	Genes tested	Full FDR05	Full FDR05 abs logFC1	Common BCV	Label permutations	MDS centroid permutation p	Control design	EQF use
GSE245885	10	5	5	20303	11766	6619	0.340	252	0.012	non-constipated tissue obtained in a colon-cancer surgery context; disease status is confounded with clinical sampling context	low-confidence contextual dataset; not confirmatory

Supplementary Table S11 | GSE245885 leave-one-sample-out sensitivity

Complete machine-readable values are retained in the repository.

Omitted sample	Omitted group	Genes tested	FDR05	FDR05 abs logFC1	LogFC spearman
NP1	NP	20303	11072	6395	0.992
NP2	NP	20303	12399	7248	0.981
NP3	NP	20303	11623	6287	0.989
NP4	NP	20303	11570	6954	0.993
NP5	NP	20303	11210	6436	0.992
STC1	STC	20303	10896	6307	0.996
STC2	STC	20303	11043	6216	0.995
STC3	STC	20303	11338	6839	0.995
STC4	STC	20303	10856	6325	0.997
STC5	STC	20303	10912	6361	0.996

Supplementary Table S12 | External formula identity audit

Complete machine-readable values are retained in the repository.

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Formula id	Formula name en	Primary n	Cross source jaccard	Etcn best jaccard	Included identity gate
F001	Si-Wu-Tang	4	1	0.667	True
F002	Suan-Zao-Ren-Tang	5	1	0.333	True
F003	Gui-Zhi-Tang	5	1	0.600	True
F004	Wen-Dan-Tang	8	1	0.750	True
F005	Shen-Ling-Bai-Zhu-San	11	0.909	0	True
F006	Sheng-Mai-San	3	1	0.750	False
F007	Xue-Fu-Zhu-Yu-Tang	11	1	1	True
F008	Si-Jun-Zi-Tang	6	0.429	0.667	False
F009	Xiao-Chai-Hu-Tang	7	1	0.625	True
F010	Bu-Zhong-Yi-Qi-Tang	10	0.800	0.667	True
F011	Chai-Hu-Shu-Gan-Tang	7	0	0	False
F012	Du-Huo-Ji-Sheng-Tang	15	1	0.562	True
F013	Liu-Wei-Di-Huang-Wan	6	1	0.750	True
F014	Ban-Xia-Xie-Xin-Tang	7	0.750	1	False
F015	Bao-He-Wan	7	0.875	0.300	True
F016	Xiang-Sha-Liu-Jun-Zi-Tang	9	0.778	0.545	False
F017	Wu-Ling-San	5	0.667	0.273	False
F018	Yi-Guan-Jian	6	1	0.500	True
F019	Tian-Wang-Bu-Xin-Dan	13	0.800	0.450	True
F020	Yang-Xin-Tang	14	0.857	0.625	True
F000	SQRCD	16	1	1	True

Supplementary Table S13 | External benchmark exclusions

Each exclusion is linked to the criterion that actually failed.

Formula ID	Formula name	Stage	Reason	Observed Jaccard
F006	Sheng-Mai-San	formula size	Formula size = 3, below the prespecified minimum of 4 herbs	1
F008	Si-Jun-Zi-Tang	identity concordance	Cross-source Jaccard = 0.429, below the prespecified threshold of 0.80	0.429
F011	Chai-Hu-Shu-Gan-Tang	identity concordance	Cross-source Jaccard = 0.000, below the prespecified threshold of 0.80	0
F014	Ban-Xia-Xie-Xin-Tang	identity concordance	Cross-source Jaccard = 0.750, below the prespecified threshold of 0.80	0.750
F016	Xiang-Sha-Liu-Jun-Zi-Tang	identity concordance	Cross-source Jaccard = 0.778, below the prespecified threshold of 0.80	0.778
F017	Wu-Ling-San	identity concordance	Cross-source Jaccard = 0.667, below the prespecified threshold of 0.80	0.667

Supplementary Table S14 | Cross-formula benchmark summary

FDI is bounded above by 0.5 under the union-coverage formulation.

Formula id	Formula name en	Formula size	Compound n nodes	Target n nodes	Compound FDI	Target FDI	Target multiherb fraction	Target FDI percentile within band
F000	SQRCD	16	2172	132	0.487	0.395	0.402	0.500
F001	Si-Wu-Tang	4	826	49	0.478	0.427	0.306	0.400
F002	Suan-Zao-Ren-Tang	5	1354	77	0.493	0.469	0.143	1
F003	Gui-Zhi-Tang	5	1093	70	0.495	0.451	0.229	0.600
F004	Wen-Dan-Tang	8	1173	104	0.495	0.440	0.269	0.500
F005	Shen-Ling-Bai-Zhu-San	11	2022	100	0.492	0.379	0.530	0.333
F007	Xue-Fu-Zhu-Yu-Tang	11	2519	126	0.485	0.425	0.341	0.833
F009	Xiao-Chai-Hu-Tang	7	1654	94	0.491	0.446	0.245	0.750
F010	Bu-Zhong-Yi-Qi-Tang	10	2582	97	0.490	0.409	0.371	0.250
F012	Du-Huo-Ji-Sheng-Tang	15	2666	115	0.482	0.371	0.478	0.167
F013	Liu-Wei-Di-Huang-Wan	6	915	69	0.493	0.405	0.420	0.200
F015	Bao-He-Wan	7	822	70	0.496	0.453	0.200	1
F018	Yi-Guan-Jian	6	728	88	0.495	0.458	0.204	0.800
F019	Tian-Wang-Bu-Xin-Dan	13	1848	115	0.494	0.410	0.365	0.667
F020	Yang-Xin-Tang	14	2673	144	0.483	0.427	0.306	1

Supplementary Table S15 | Exploratory HC3 coefficients (n = 15)

This descriptive model is a metric-construction check. Extremely small P values primarily reflect the mathematical dependence between FDI and cross-herb redundancy and are not independent biological evidence.

Node type	Term	Estimate	HC3 SE	P value	95% CI lower	95% CI upper
compound	Intercept	0.501	0.004	< 0.001	0.493	0.509
compound	Formula size	0.000	0.000	0.435	-0.000	0.000
compound	log(1 + number of nodes)	-0.000	0.001	0.771	-0.001	0.001
compound	Cross-herb redundancy	-0.216	0.015	3.68×10^{-49}	-0.245	-0.188
target	Intercept	0.397	0.033	1.07×10^{-32}	0.332	0.462
target	Formula size	-0.003	0.001	4.63×10^{-5}	-0.005	-0.002
target	log(1 + number of nodes)	0.028	0.008	6.72×10^{-4}	0.012	0.044
target	Cross-herb redundancy	-0.216	0.016	1.30×10^{-40}	-0.247	-0.184

Supplementary Table S16 | Unified eight-domain state profiles

The benchmark assesses identity and topology portability; unavailable downstream evidence is not imputed.

Formula id	Formula name en	Formula identity	Preparation chemistry	Chemical reconstruction	Target provenance	Human disease context	Human genetics	Formula perturbation and task identifiability	Structural eligibility
F000	SQRCD	qualified	partially qualified	partially qualified	partially qualified	partially qualified	not estimable	not estimable	stopped

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Formula id	Formula name en	Formula identity	Preparation chemistry	Chemical reconstruction	Target provenance	Human disease context	Human genetics	Formula perturbation and task identifiability	Structural eligibility
F001	Si-Wu-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F002	Suan-Zao-Ren-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F003	Gui-Zhi-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F004	Wen-Dan-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F005	Shen-Ling-Bai-Zhu-San	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F007	Xue-Fu-Zhu-Yu-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F009	Xiao-Chai-Hu-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F010	Bu-Zhong-Yi-Qi-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F012	Du-Huo-Ji-Sheng-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F013	Liu-Wei-Di-Huang-Wan	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F015	Bao-He-Wan	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F018	Yi-Guan-Jian	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F019	Tian-Wang-Bu-Xin-Dan	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped
F020	Yang-Xin-Tang	qualified	not estimable	partially qualified	partially qualified	not estimable	not estimable	not estimable	stopped

Supplementary Table S17 | Compound entity-harmonization sensitivity

PubChem CID and parent InChIKey analyses were not estimable because uniform identifiers were absent.

Formula id	Formula name en	Native nodes	Normalized name nodes	Native FDI	Normalized name FDI	FDI change x1000
F000	SQRCD	2172	2106	0.488	0.487	0.517
F001	Si-Wu-Tang	826	803	0.478	0.478	0.373
F002	Suan-Zao-Ren-Tang	1354	1332	0.493	0.493	0.116
F003	Gui-Zhi-Tang	1093	1069	0.495	0.495	0.110
F004	Wen-Dan-Tang	1173	1152	0.495	0.495	0.092
F005	Shen-Ling-Bai-Zhu-San	2022	1966	0.493	0.492	0.121
F007	Xue-Fu-Zhu-Yu-Tang	2519	2453	0.485	0.485	0.174
F009	Xiao-Chai-Hu-Tang	1654	1626	0.491	0.490	0.395
F010	Bu-Zhong-Yi-Qi-Tang	2582	2522	0.490	0.490	0.232
F012	Du-Huo-Ji-Sheng-Tang	2666	2597	0.482	0.482	0.383
F013	Liu-Wei-Di-Huang-Wan	915	891	0.493	0.493	0.185
F015	Bao-He-Wan	822	791	0.496	0.496	0.150
F018	Yi-Guan-Jian	728	699	0.495	0.495	0.214
F019	Tian-Wang-Bu-Xin-Dan	1848	1771	0.494	0.494	0.214
F020	Yang-Xin-Tang	2673	2601	0.483	0.482	0.377

Supplementary Table S18 | Returned rater-workbook quality audit

The blank template was excluded; identifying fields are not reproduced.

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File role	Rows	Formulae	Rubric items	Completed ratings	Completion rate	Invalid state rows	Usable for agreement	Finding
blank template	750	15	50	0	0	0	False	Template structure is complete, but all 750 rating, initials and date cells are blank; not usable as a rater record.
completed rating A	750	15	50	750	1	0	True	Complete 750-item record with unique formula-item keys and allowed states.
completed rating B	750	15	50	750	1	0	True	Complete 750-item record with unique formula-item keys and allowed states.

Supplementary Table S19 | Overall field-level agreement

Observed four-state results are primary; collapsed and rule-constrained outputs are sensitivity analyses.

Analysis	N	Agreement n	Raw agreement	Cohen κ unweighted	Interpretation
original four state	750	689	0.919	0.847	Observed ratings; no post hoc adjudication.
collapsed unavailable sensitivity	750	749	0.999	0.997	Analytical collapse of two unavailable process states; not a re-rating.
rule constrained structural sensitivity	750	749	0.999	0.998	Deterministic application of the clarified structural stopping rule; not assessor re-rating.

Supplementary Table S20 | Domain-specific agreement

NE*: Cohen κ is not estimable because both assessors assigned a constant category. Structural eligibility had 0/60 agreement under the original rubric.

Domain	Analysis	N	Agreement, n	Raw agreement	Unweighted Cohen κ
Formula identity	original four state	150	150	1	1
Formula identity	collapsed unavailable sensitivity	150	150	1	1
Formula identity	rule constrained structural sensitivity	150	150	1	1
Preparation chemistry	original four state	105	105	1	1
Preparation chemistry	collapsed unavailable sensitivity	105	105	1	1
Preparation chemistry	rule constrained structural sensitivity	105	105	1	1
Chemical reconstruction	original four state	90	90	1	1
Chemical reconstruction	collapsed unavailable sensitivity	90	90	1	1
Chemical reconstruction	rule constrained structural sensitivity	90	90	1	1
Target provenance	original four state	120	120	1	1

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Domain	Analysis	N	Agreement, n	Raw agreement	Unweighted Cohen κ
Target provenance	collapsed unavailable sensitivity	120	120	1	1
Target provenance	rule constrained structural sensitivity	120	120	1	1
Human disease context	original four state	90	90	1	1
Human disease context	collapsed unavailable sensitivity	90	90	1	1
Human disease context	rule constrained structural sensitivity	90	90	1	1
Human genetics	original four state	75	75	1	NE*
Human genetics	collapsed unavailable sensitivity	75	75	1	NE*
Human genetics	rule constrained structural sensitivity	75	75	1	NE*
Formula perturbation and task identifiability	original four state	60	59	0.983	0.496
Formula perturbation and task identifiability	collapsed unavailable sensitivity	60	59	0.983	0.496
Formula perturbation and task identifiability	rule constrained structural sensitivity	60	59	0.983	0.496
Structural eligibility	original four state	60	0	0	0
Structural eligibility	collapsed unavailable sensitivity	60	60	1	NE*
Structural eligibility	rule constrained structural sensitivity	60	60	1	NE*

Supplementary Table S21 | Formula-specific agreement

Formula-level results expose the repeated structural-state boundary.

Formula id	Analysis	N items	Agreement n	Raw agreement	Cohen κ unweighted
F000	original four state	50	45	0.900	0.841
F000	collapsed unavailable sensitivity	50	49	0.980	0.967
F000	rule constrained structural sensitivity	50	49	0.980	0.969
F001	original four state	50	46	0.920	0.844
F001	collapsed unavailable sensitivity	50	50	1	1
F001	rule constrained structural sensitivity	50	50	1	1
F002	original four state	50	46	0.920	0.844
F002	collapsed unavailable sensitivity	50	50	1	1
F002	rule constrained structural sensitivity	50	50	1	1
F003	original four state	50	46	0.920	0.844
F003	collapsed unavailable sensitivity	50	50	1	1
F003	rule constrained structural sensitivity	50	50	1	1
F004	original four state	50	46	0.920	0.844
F004	collapsed unavailable sensitivity	50	50	1	1
F004	rule constrained structural sensitivity	50	50	1	1
F005	original four state	50	46	0.920	0.844
F005	collapsed unavailable sensitivity	50	50	1	1
F005	rule constrained structural sensitivity	50	50	1	1
F007	original four state	50	46	0.920	0.844
F007	collapsed unavailable sensitivity	50	50	1	1

EQF | Supplementary Information

Formula id	Analysis	N items	Agreement n	Raw agreement	Cohen κ unweighted
F007	rule constrained structural sensitivity	50	50	1	1
F009	original four state	50	46	0.920	0.844
F009	collapsed unavailable sensitivity	50	50	1	1
F009	rule constrained structural sensitivity	50	50	1	1
F010	original four state	50	46	0.920	0.844
F010	collapsed unavailable sensitivity	50	50	1	1
F010	rule constrained structural sensitivity	50	50	1	1
F012	original four state	50	46	0.920	0.844
F012	collapsed unavailable sensitivity	50	50	1	1
F012	rule constrained structural sensitivity	50	50	1	1
F013	original four state	50	46	0.920	0.844
F013	collapsed unavailable sensitivity	50	50	1	1
F013	rule constrained structural sensitivity	50	50	1	1
F015	original four state	50	46	0.920	0.844
F015	collapsed unavailable sensitivity	50	50	1	1
F015	rule constrained structural sensitivity	50	50	1	1
F018	original four state	50	46	0.920	0.844
F018	collapsed unavailable sensitivity	50	50	1	1
F018	rule constrained structural sensitivity	50	50	1	1
F019	original four state	50	46	0.920	0.844
F019	collapsed unavailable sensitivity	50	50	1	1
F019	rule constrained structural sensitivity	50	50	1	1
F020	original four state	50	46	0.920	0.844
F020	collapsed unavailable sensitivity	50	50	1	1
F020	rule constrained structural sensitivity	50	50	1	1

Supplementary Table S22 | Leave-one-formula-out κ sensitivity

Clusters are formulae; the narrow range reflects repeated rating structure.

Omitted formula id	Analysis	N	Raw agreement	Cohen κ unweighted
F000	original four state	700	0.920	0.844
F000	collapsed unavailable sensitivity	700	1	1
F000	rule constrained structural sensitivity	700	1	1
F001	original four state	700	0.919	0.847
F001	collapsed unavailable sensitivity	700	0.999	0.997
F001	rule constrained structural sensitivity	700	0.999	0.998
F002	original four state	700	0.919	0.847
F002	collapsed unavailable sensitivity	700	0.999	0.997
F002	rule constrained structural sensitivity	700	0.999	0.998
F003	original four state	700	0.919	0.847
F003	collapsed unavailable sensitivity	700	0.999	0.997
F003	rule constrained structural sensitivity	700	0.999	0.998

EQF | Supplementary Information

Omitted formula id	Analysis	N	Raw agreement	Cohen κ unweighted
F004	original four state	700	0.919	0.847
F004	collapsed unavailable sensitivity	700	0.999	0.997
F004	rule constrained structural sensitivity	700	0.999	0.998
F005	original four state	700	0.919	0.847
F005	collapsed unavailable sensitivity	700	0.999	0.997
F005	rule constrained structural sensitivity	700	0.999	0.998
F007	original four state	700	0.919	0.847
F007	collapsed unavailable sensitivity	700	0.999	0.997
F007	rule constrained structural sensitivity	700	0.999	0.998
F009	original four state	700	0.919	0.847
F009	collapsed unavailable sensitivity	700	0.999	0.997
F009	rule constrained structural sensitivity	700	0.999	0.998
F010	original four state	700	0.919	0.847
F010	collapsed unavailable sensitivity	700	0.999	0.997
F010	rule constrained structural sensitivity	700	0.999	0.998
F012	original four state	700	0.919	0.847
F012	collapsed unavailable sensitivity	700	0.999	0.997
F012	rule constrained structural sensitivity	700	0.999	0.998
F013	original four state	700	0.919	0.847
F013	collapsed unavailable sensitivity	700	0.999	0.997
F013	rule constrained structural sensitivity	700	0.999	0.998
F015	original four state	700	0.919	0.847
F015	collapsed unavailable sensitivity	700	0.999	0.997
F015	rule constrained structural sensitivity	700	0.999	0.998
F018	original four state	700	0.919	0.847
F018	collapsed unavailable sensitivity	700	0.999	0.997
F018	rule constrained structural sensitivity	700	0.999	0.998
F019	original four state	700	0.919	0.847
F019	collapsed unavailable sensitivity	700	0.999	0.997
F019	rule constrained structural sensitivity	700	0.999	0.998
F020	original four state	700	0.919	0.847
F020	collapsed unavailable sensitivity	700	0.999	0.997
F020	rule constrained structural sensitivity	700	0.999	0.998

Supplementary Table S23 | SQRCD rank within the size band

Rank and denominator are reported together.

Formula id	Comparison band	N in band	Rank low to high FDI	Rank high to low FDI	Percentile within band	Target FDI
F000	11-20	6	3	4	0.500	0.395

Supplementary Table S24 | Inter-rater disagreement taxonomy

No disagreement was hidden by post hoc adjudication.

Disagreement type	N	Share of disagreements	Interpretation
stopped versus not estimable	60	98.4%	Systematic downstream-state definition boundary across structural fields

Disagreement type	N	Share of disagreements	Interpretation
partially qualified versus qualified	1	1.6%	Single SQRCD PE04 evidence-threshold disagreement

Supplementary Table S25 | Independent-assessor procedure record

Organizer-reported and protocol-specified information is distinguished from undocumented information.

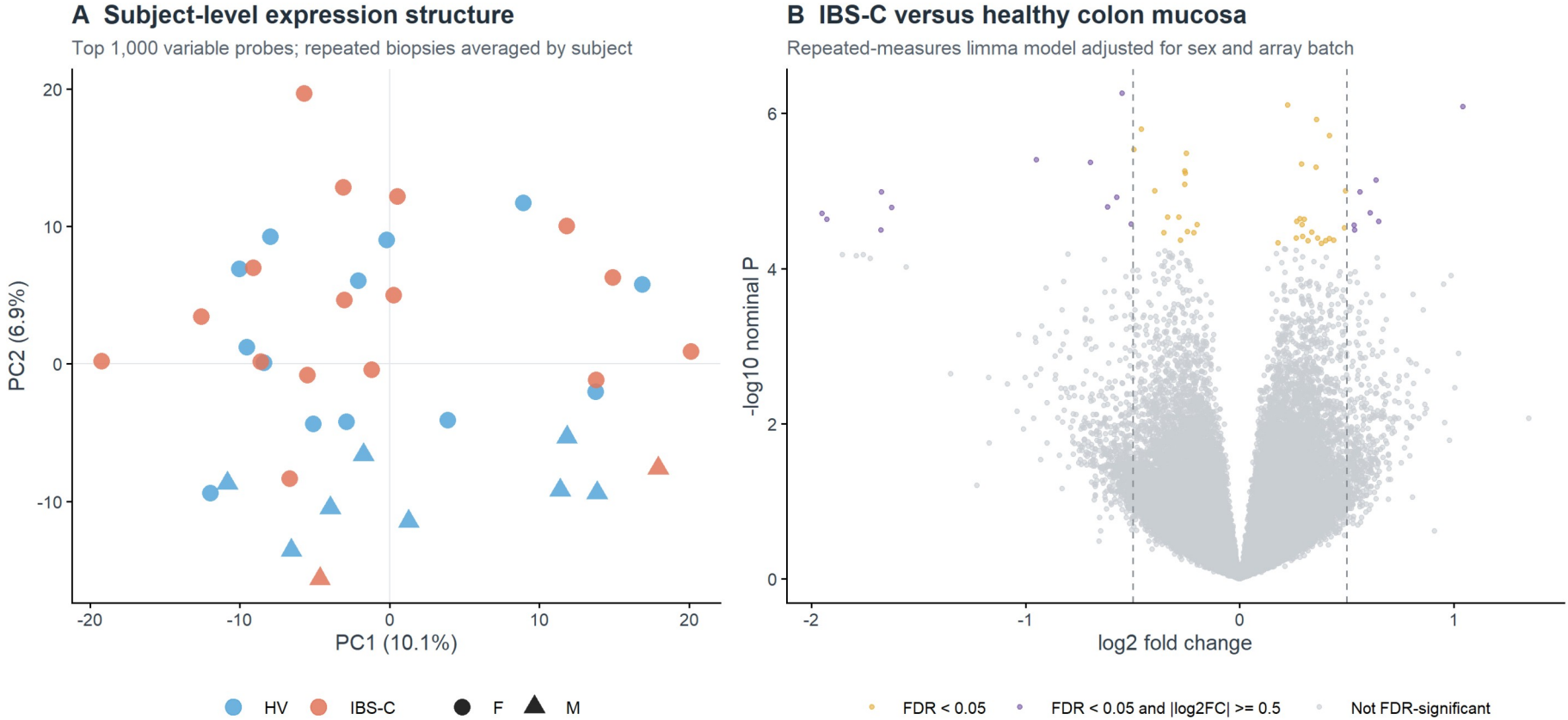
Field	Record	Status
Assessor backgrounds	Both assessors had medical backgrounds and working knowledge of network pharmacology.	organizer reported
Relationship to author team	Neither assessor was a member of the author team.	organizer reported
Role in EQF design	Neither assessor participated in EQF design.	organizer reported
Training	Assessment-only orientation to the locked manual and rubric; no study-conclusion briefing.	organizer reported
Evidence access	Both assessors received the same locked evidence-reference table and dossiers; ratings and comments were entered independently.	organizer reported
Access restrictions	The second assessor had no access to the first assessor's ratings, agreement outputs or final manuscript domain states before workbook submission.	organizer reported
External searching	Not permitted by the assessment manual; dossier evidence was the only scoring source.	protocol specified
Rating duration	One assessment took approximately half a day and the other approximately one day; assessor-specific mapping was not retained.	organizer reported
Pre-scoring discussion	No auditable record was retained; absence of discussion cannot be verified from workbook metadata.	not documented
Workbook freeze	Completed workbooks were frozen on their recorded submission dates (2026-08-01 and 2026-08-02) before agreement analysis.	workbook metadata plus organizer record

Supplementary Table S26 | Stopped versus not estimable decision rules

The clarified rule is prospective; observed ratings remain unchanged.

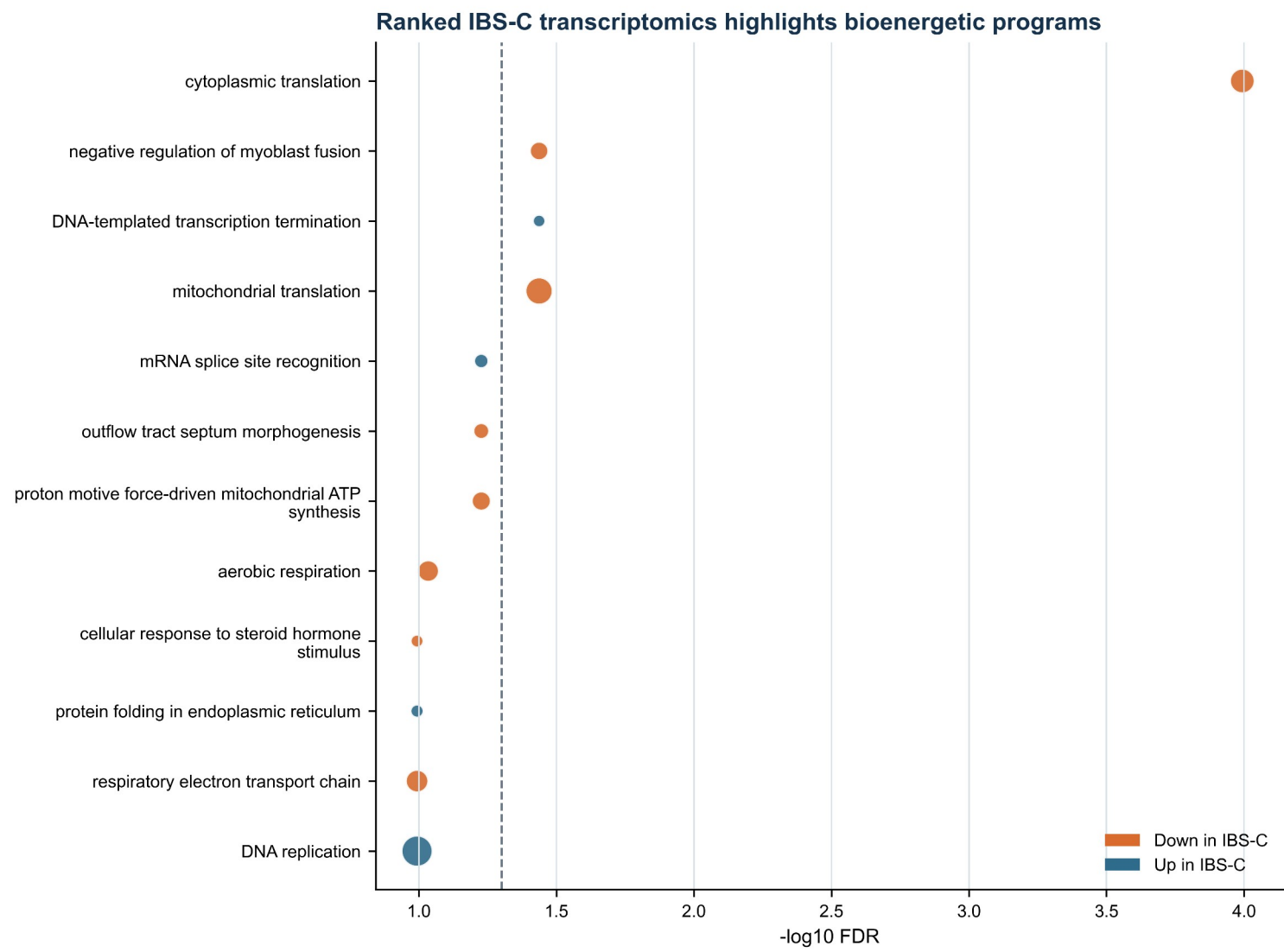
Question	Condition	State	Operational rule
Is the required input absent?	Yes, but no prespecified downstream halt applies	not estimable	Record the missing input; do not infer a negative result.
Did an upstream mandatory gate fail?	Yes, and the protocol prohibits downstream continuation	stopped	Apply stopped; the assessor may not substitute not estimable.
Does structural eligibility survive upstream gates?	No eligible compound-protein pair survives preparation relevance, target provenance and human context	stopped	Docking and molecular dynamics are withheld.

Supplementary Figure S1 | GSE36701 diagnostic panels



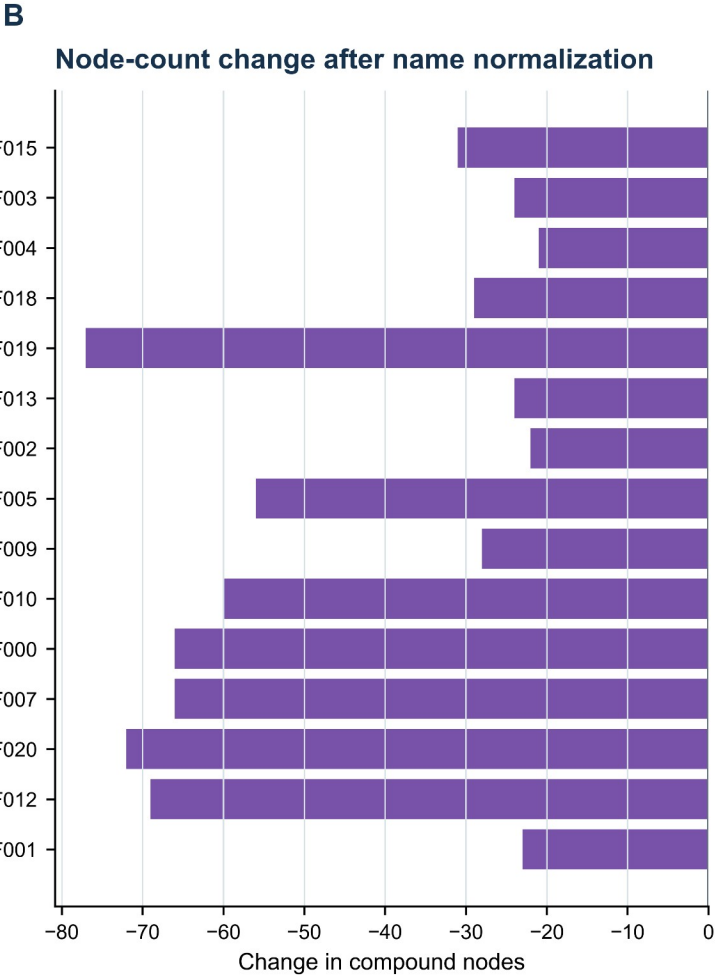
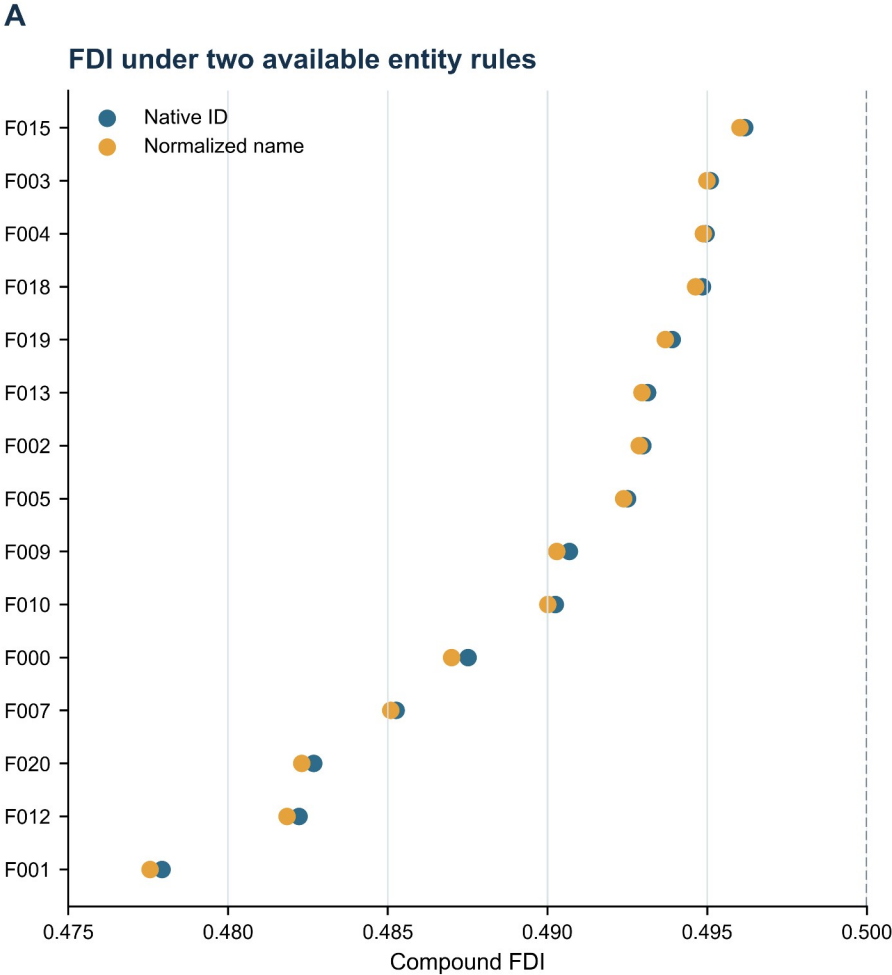
Participant-aware rectal mucosal diagnostics provide disease context, not a formula response.

Supplementary Figure S2 | GSE36701 ranked pathway context



Full pathway names are wrapped rather than truncated; ranked pathways are supportive disease context only.

Supplementary Figure S3 | Compound entity-harmonization sensitivity

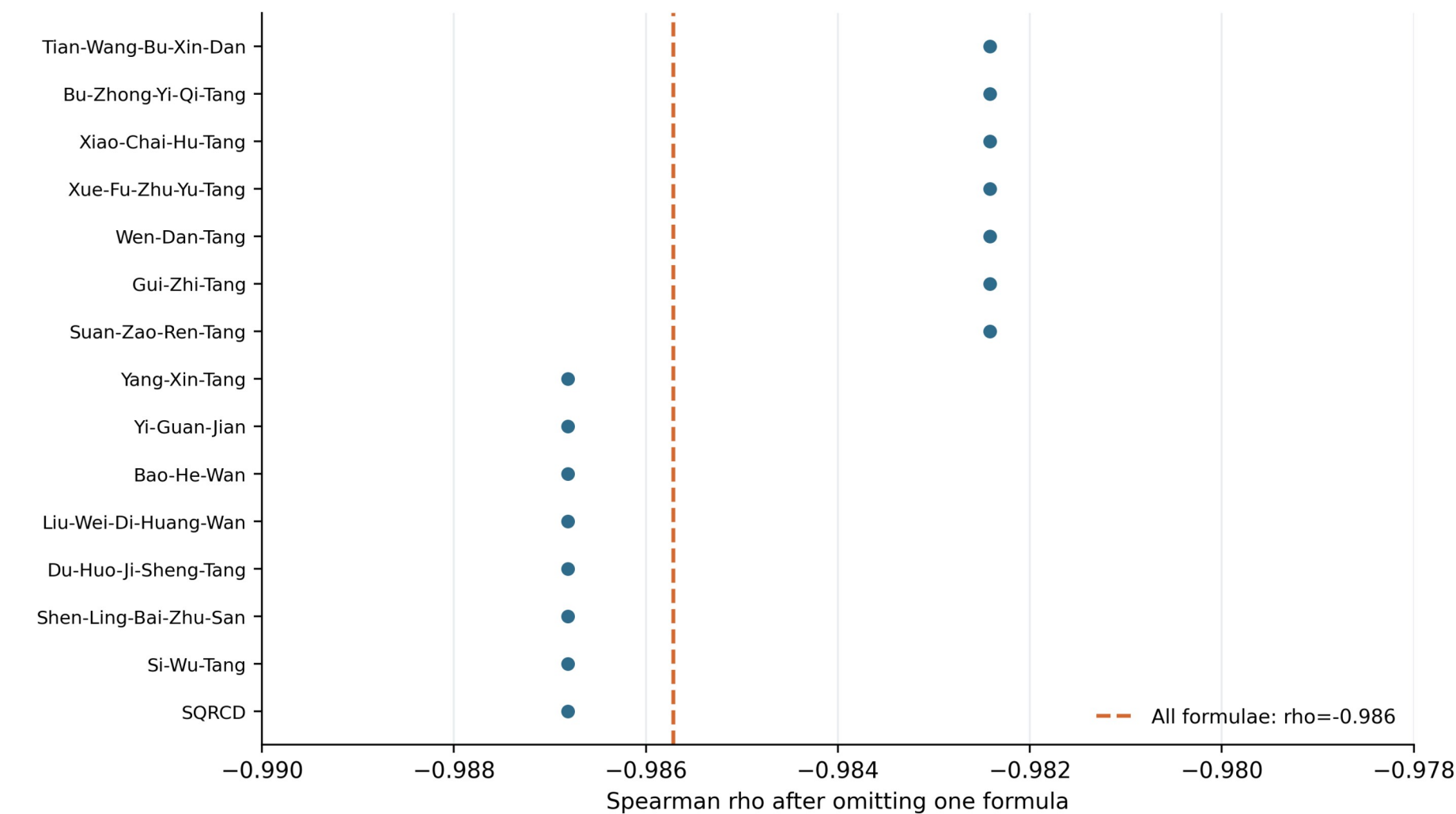


FDI and node counts were compared across the two uniformly available entity representations. PubChem CID and parent InChIKey were not uniformly available.

Supplementary Figure S4 | Leave-one-formula-out FDI sensitivity

Leave-one-formula-out metric-behavior sensitivity

15 omissions; the redundancy-FDI relation remains definition-linked

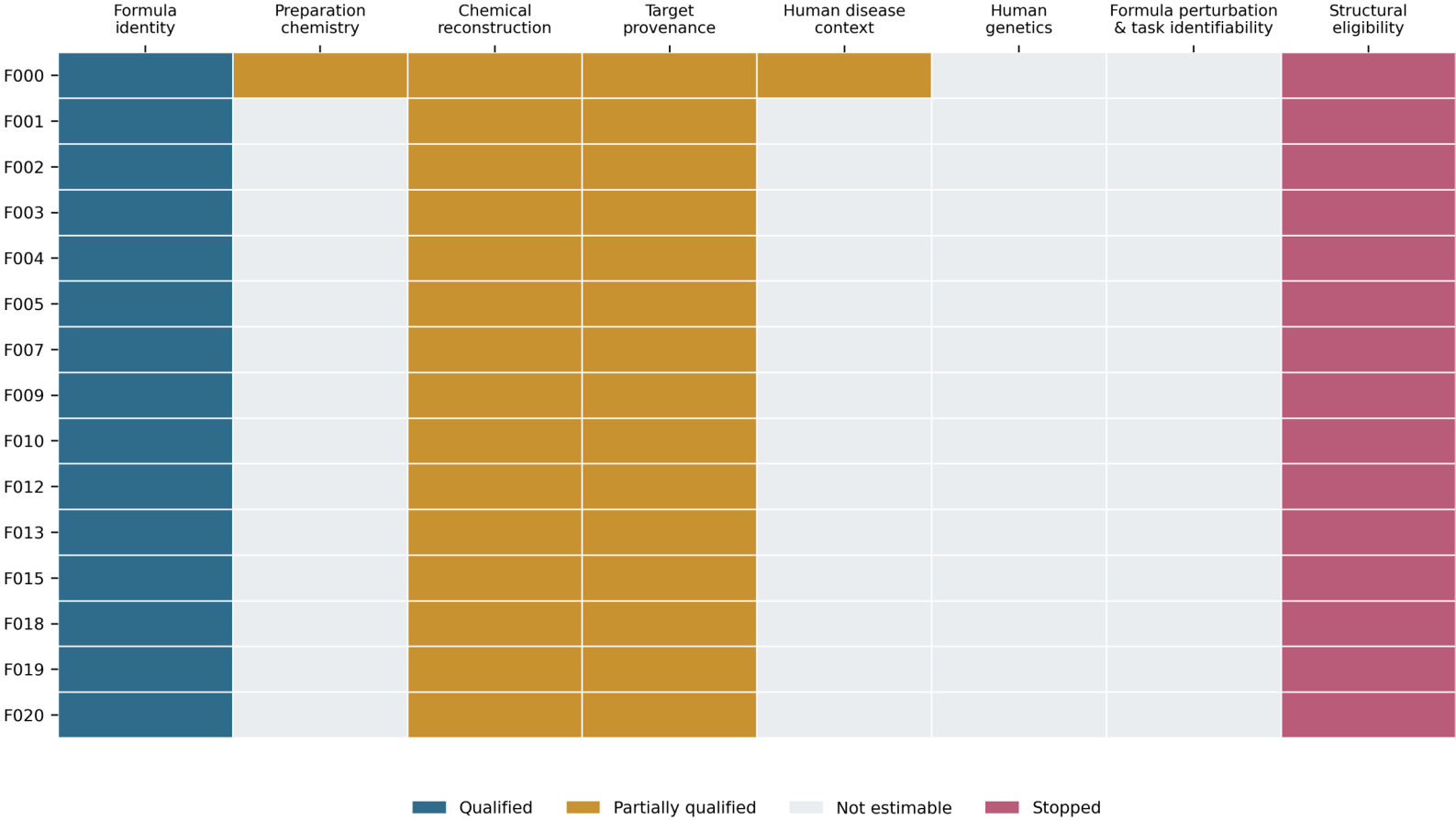


The redundancy-FDI relation remained construction-linked after each formula omission.

Supplementary Figure S5 | Unified domain-state portability

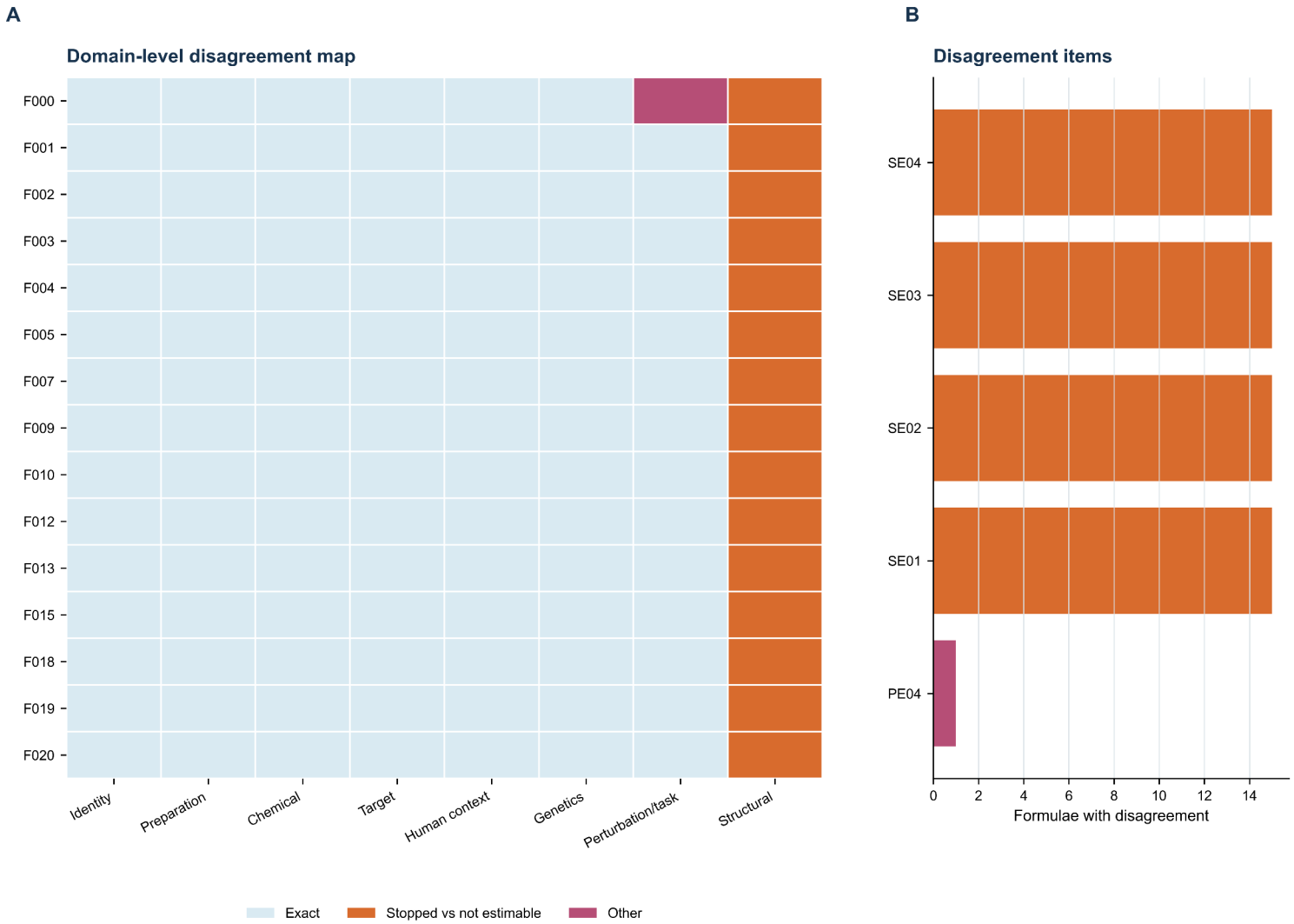
Unified eight-domain EQF state profiles

The benchmark evaluates identity and topology portability; downstream domains remain unavailable or stopped.



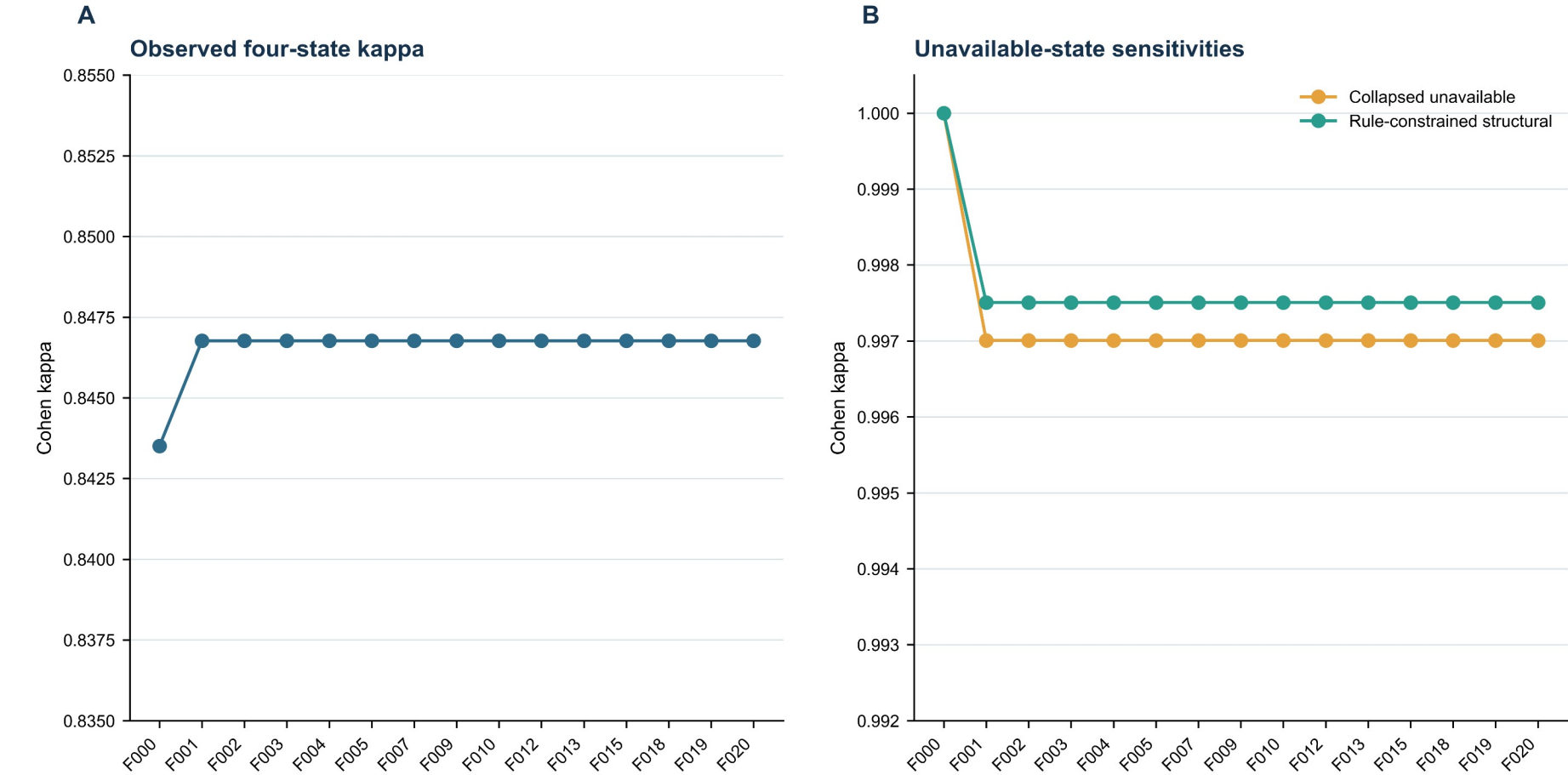
All 15 formulae are shown against the same eight-domain vocabulary.

Supplementary Figure S6 | Domain-level inter-rater disagreement



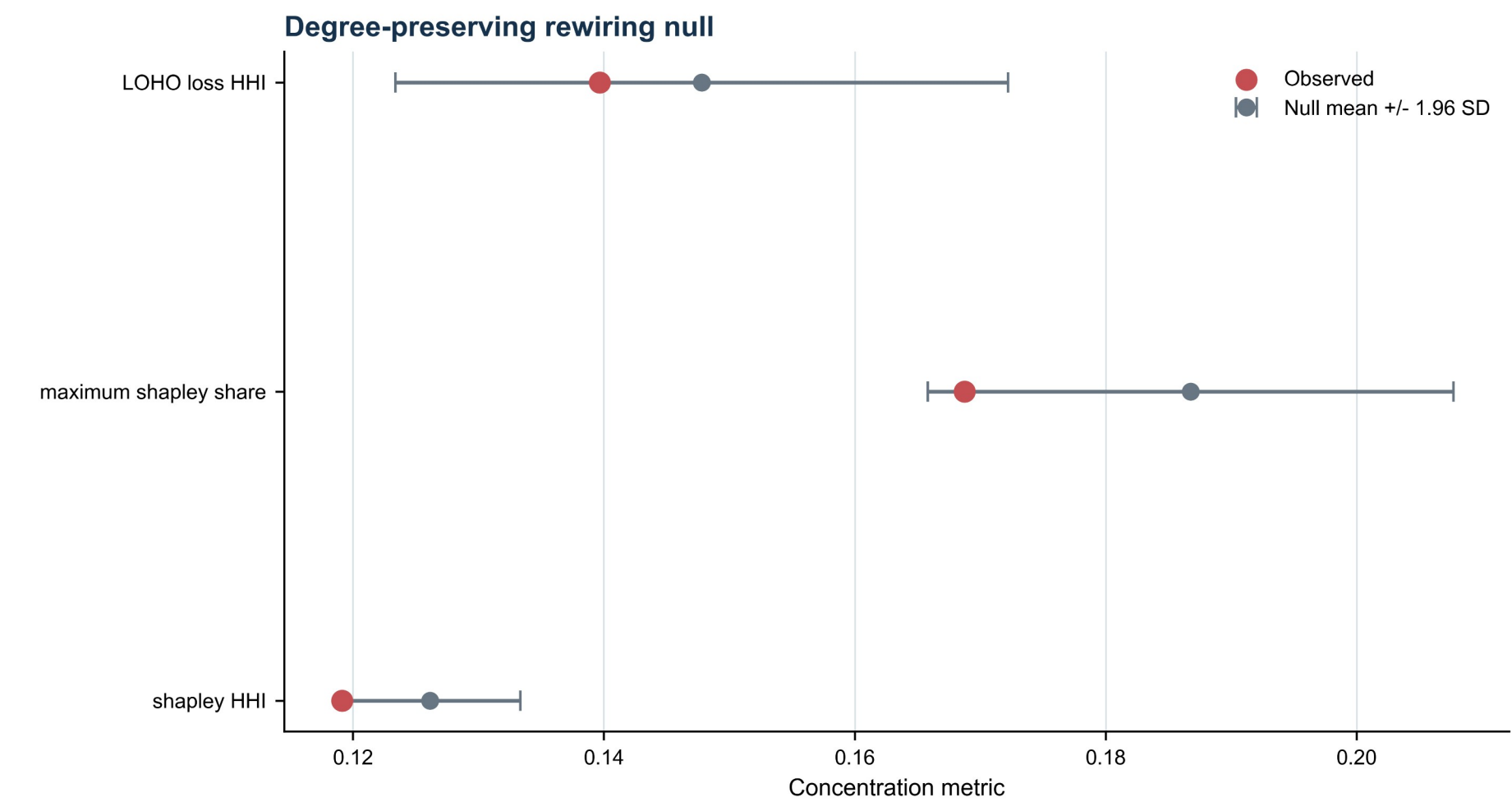
Domain-level aggregation replaces 50 unreadable x-axis labels; the right panel identifies the five item IDs containing disagreements.

Supplementary Figure S7 | Leave-one-formula-out κ sensitivity



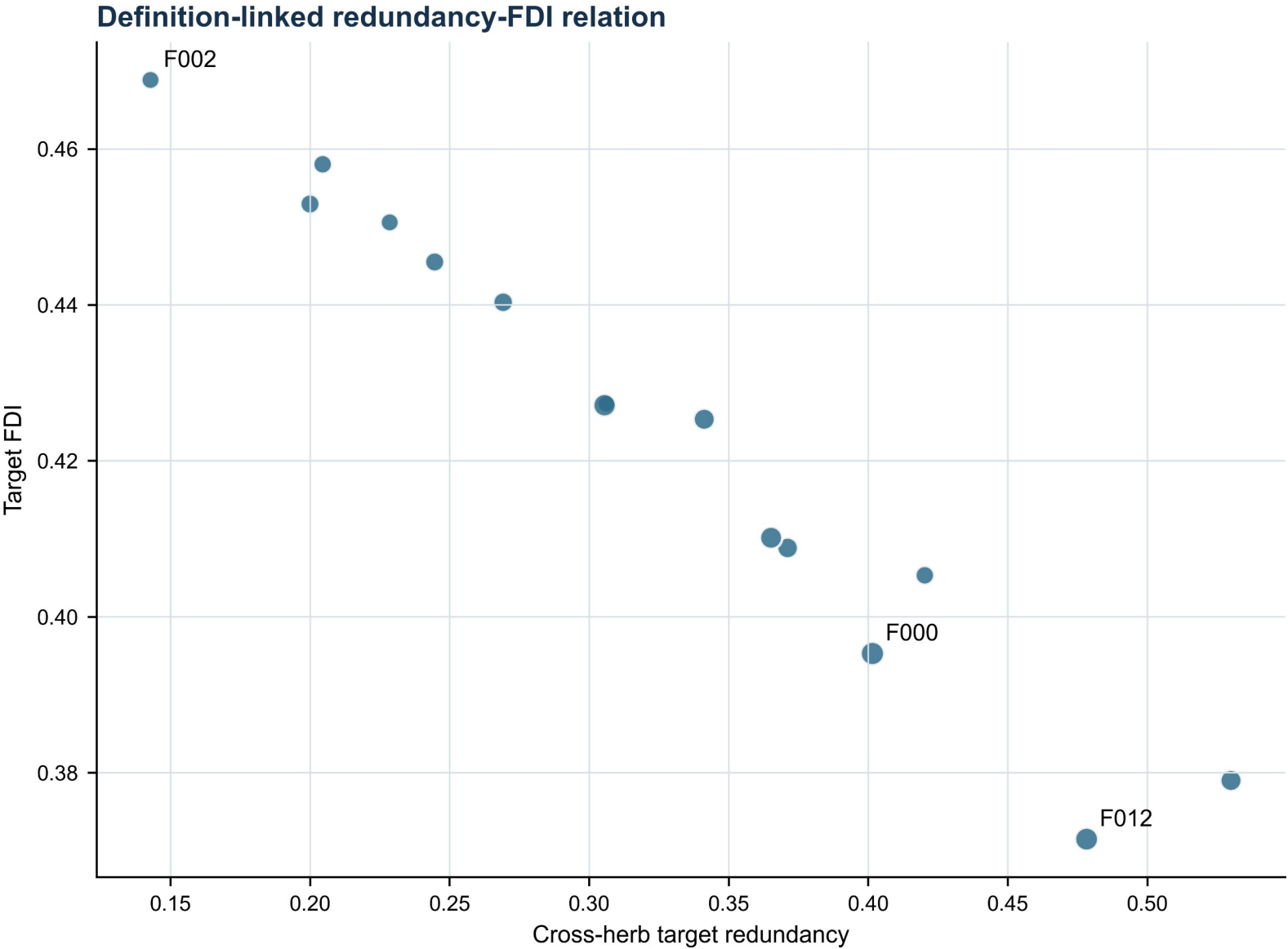
Observed four-state κ and the two sensitivity analyses are separated onto appropriate y-axis ranges.

Supplementary Figure S8 | Degree-preserving rewiring null



Observed concentration metrics are compared with degree-preserving null means and 1.96 standard deviations.

Supplementary Figure S9 | Definition-linked redundancy-FDI relation



Labels identify the minimum, maximum and SQRC; the association is a metric-behaviour check rather than independent biological evidence.